Minhaj Uddin Ahmad

+1-847-262-8112 | mahmad12@crimson.ua.edu | minhaj6.github.io

Minhaj6 | Minhaj Uddin Ahmad

Tuscaloosa, Alabama - 35404

EDUCATION

- **University of Alabama** *January 2024 - Present*
  Ph.D. Civil Engineering Tuscaloosa, USA
- **University of Alabama** *January 2024 - December 2025*
  MSc. Civil Engineering Tuscaloosa, USA
  - GPA: 4.00/4.00
- **Islamic University of Technology** *May 2023*
  Bsc. Electrical Engineering Dhaka, Bangladesh
  - GPA: 3.78/4.00

EXPERIENCE

- **University of Alabama [🌐]** *January 2024 - Present*
  Graduate Research Assistant Tuscaloosa, AL
  - Developed OpenCAMS, a co-simulation platform integrating traffic simulation, sensor simulation and C-V2X communications simulation by combining SUMO, CARLA and OMNeT++ simulators.
  - Implemented late-fusion of roadside LiDAR, Radar, Camera for proactive safety warning system.
  - Worked on implementation of GNSS spoofing detection and mitigation techniques, demonstrated on TraCR Annual Conference 2024 and USDOT FoT Summit 2024.
- **Cybernetics Hi Tech Solutions (Pvt) Ltd. [🌐]** *May 2023 - December 2023*
  Research Engineer (Embedded Systems) Dhaka, Bangladesh
  - Developed live multi-camera low-latency video streaming over radio transceivers from deployed robotic platforms.
  - Designed CAN-BUS data streaming module over Ethernet interface on STM32-H7 microcontroller for controlling robotic platforms.
- **ANTS Aerial Systems Ltd. [🌐]** *2022 - 2023*
  Engineering Intern Dhaka, Bangladesh
  - Worked on drone swarm flight firmware, based on Ardupilot flight controller firmware.
  - Worked on software-in-the-loop (SITL) simulation of automated mission-planning software, over an implementation of MAVLink protocol.

PRESENTATIONS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, B=BOOK CHAPTER

Journal Papers

[J.3] **Ahmad, M.U.**, Dasgupta, S., Rahman, M., Khan, S., Haque, M.W., Saba, S.R., Bodoh, D., Huynh, N., Zhao, L., and Ozguven, E.E. (2026). **Lessons Learned from the Real-World Deployment of Multi-Sensor Fusion for Proactive Work Zone Safety Application**. *Transportation Research Record*. DOI: 10.1177/03611981261441281

[J.2] Saba, S.R., Khan, S., **Ahmad, M.U.**, Cao, J., Dasgupta, S., Rahman, M., Haque, M.W., Bodoh, D., Zhao, L., Huynh, N., and Ozguven, E.E. (2026). **Infrastructure Sensor-Enabled Vehicle Trajectory Generation Using Multimodal Fusion for Proactive Safety Applications at Work Zone**. *Transportation Research Record*. (Accepted for publication)

[J.1] **Ahmad, M.U.**, Rahman, M., Sevim, A., Bodoh, D., Khan, S., Zhao, L., Huynh, N., and Ozguven, E.E. (2026). **Historical Prediction Attention Mechanism Based Trajectory Forecasting for Proactive Work Zone Safety in a Digital Twin Environment**. *IEEE Internet of Things Journal*, Vol. 13, Issue 11, pp. 22846–22861. DOI: 10.1109/JIOT.2026.3669235

Book Chapters

[B.2] **Ahmad, M.U.**, Abrar, A., Dasgupta, S., and Rahman, M. (2026). **End-to-End Co-Simulation Testbed for Cybersecurity Research and Development in Intelligent Transportation Systems**. In *Advances in Transportation Cybersecurity and Resilience*, World Scientific Publishing. DOI: 10.1142/9789819831548_0006

[B.1] Rahman, M., **Ahmad, M.U.**, Chowdhury, M., and McClendon, J. (2025). **Data Analytics for Prediction of Real-Time Traffic Flow Parameters**. In *Data Analytics for Intelligent Transportation Systems*, pp. 289–315, Elsevier. DOI: 10.1016/B978-0-443-13878-2.00011-4

Peer-reviewed Conference Proceedings and Presentations

[C.18] **Ahmad, M.U.**, Bhuiyan, M.Z.H., Irfan, M.S., Dasgupta, S., and Rahman, M. (2027). **Geometry- and Timing-Aware Multi-Constellation PVT Cluster Matching for GNSS Spoofing Detection and Mitigation**. In *2027 IEEE/ION Position, Location and Navigation Symposium (PLANS 2027), B6b: Integrity and Augmentation 2, Portland, Oregon, USA, Apr 12–15, 2027*. Accepted.

[C.17] **Ahmad, M.U.**, Irfan, M.S., Dasgupta, S., Bandi, T.N., Rahman, M., and Chowdhury, M. (2027). **Precision Timing-Based Spoofing Detection and Integrity-Aware Correction Dissemination for Resilient RTK Infrastructure**. In *2027 IEEE/ION Position, Location and Navigation Symposium (PLANS 2027), A3a: Inertial Sensors, Calibration and Precision Timing, Portland, Oregon, USA, Apr 12–15, 2027*. Accepted.

[C.16] **Ahmad, M.U.**, Castillo Agüero, V., Irfan, M.S., Gómez López, M.Á., Dasgupta, S., and Rahman, M. (2027). **Forging SBAS Messages Against Drone Operations: Threat Model, Attack Formulation and Evaluation with Authentic GPS Data**. In *2027 IEEE/ION Position, Location and Navigation Symposium (PLANS 2027), B6b: Integrity and Augmentation 2, Portland, Oregon, USA, Apr 12–15, 2027*. Accepted.

[C.15] Irfan, M.S., Haque, M.W., **Ahmad, M.U.**, Dasgupta, S., and Rahman, M. (2027). **LiDAR–Inertial Global Localization Using OpenStreetMap Geometry for GNSS-Denied Autonomous Vehicle Navigation**. In *2027 IEEE/ION Position, Location and Navigation Symposium (PLANS 2027), A5: Inertial Navigation and Multisensor Integration, Portland, Oregon, USA, Apr 12–15, 2027*. Accepted.

[C.14] Castillo Agüero, V., **Ahmad, M.U.**, Rahman, M., and Gómez López, M.Á. (2027). **Local SBAS Integrity for Drone Operations: Decomposing EGNOS Protection Levels Along a Real Flight with an Open-Source SDR Receiver**. In *2027 IEEE/ION Position, Location and Navigation Symposium (PLANS 2027), B2a: Integrity and Augmentation 1, Portland, Oregon, USA, Apr 12–15, 2027*. Accepted.

[C.13] Abrar, A., Dasgupta, S., Mamun, A.A., **Ahmad, M.U.**, Rahman, M., Chowdhury, M., and Alsharif, A. (2027). **Deployment Feasibility Analysis of Post-Quantum Digital Signatures in Safety-Critical C-V2X Communication for Urban Mobility Scenario**. In *Transportation Research Board Annual Meeting, Washington, DC, USA, Jan 10–14, 2027*.

[C.12] **Ahmad, M.U.**, Dasgupta, S., Irfan, M.S., Rahman, M., Chowdhury, M., and Bandi, T.N. (2027). **Treating Statewide CORS Networks as Spatially Distributed Sensors for GNSS Integrity Monitoring under Unintentional and Deliberate Threats**. In *Transportation Research Board Annual Meeting, Washington, DC, USA, Jan 10–14, 2027*.

[C.11] Kong, J., **Ahmad, M.U.**, Hainen, A., Rahman, M., and Jones, S. (2027). **Evaluating Roadside Responder Safety Using Camera and LiDAR: Driver Behavior Analysis and Real-Time Near-Miss Detection**. In *Transportation Research Board Annual Meeting, Washington, DC, USA, Jan 10–14, 2027*.

[C.10] **Ahmad, M.U.**, Irfan, M.S., Dasgupta, S., and Rahman, M. (2026). **gr-pocketsdr: Bringing PocketSDR Front-Ends to GNU Radio and GNSS-SDR Ecosystem**. In *GNU Radio Conference 2026, Hardware for GNU Radio Track, Raleigh, North Carolina, USA, Sep 21–25, 2026*. Accepted for technical proceedings and talk.

[C.9] **Ahmad, M.U.**, Zaman, Z., Sun, S., and Rahman, M. (2026). **Small Language Model-Enabled Autonomous Agent for Language-Conditioned Cognitive Radar**. In *2026 IEEE International Workshop on Machine Learning for Signal Processing, Atlanta, Georgia, USA, Sep 28–Oct 1, 2026*.

[C.8] **Ahmad, M.U.**, Irfan, M.S., Dasgupta, S., and Rahman, M. (2026). **Beyond Legacy L1 C/A: Characterizing Advanced Multi-Frequency Spoofing Against Multi-Frequency GPS Receivers for Autonomous Navigation**. In *The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), C1: Navigation Resilience to Interference and Cyber-Attacks 1, Orlando, Florida, USA, Sep 14–18, 2026*. DOI: 10.33012/2026.20673

[C.7] Dasgupta, S., **Ahmad, M.U.**, Irfan, M.S., Rahman, M., and Larson, J.D. (2026). **Resilient Urban Vehicle Navigation in GNSS-Denied Environments via Tightly Integrated Inertial Propagation, Deep Motion Inference, GIS Route Constraints, and Opportunistic Visual Landmark Corrections**. In *The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), C1: Navigation Resilience to Interference and Cyber-Attacks 1, Orlando, Florida, USA, Sep 14–18, 2026*. DOI: 10.33012/2026.20799

[C.6] Irfan, M.S., **Ahmad, M.U.**, Zaman, Z., Dasgupta, S., and Rahman, M. (2026). **A Synchronous GNSS Spoofing Attack Model by Exploiting Basic Safety Message Transmissions for Connected and Automated Vehicles**. In *The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), F3: Resilient and Secure PNT Against Interference and Emerging Threats 2, Orlando, Florida, USA, Sep 14–18, 2026*. DOI: 10.33012/2026.20651

[C.5] **Ahmad, M.U.**, Dasgupta, S., Rahman, M., Khan, S., Haque, M.W., Saba, S.R., Bodoh, D., Huynh, N., Zhao, L., and Ozguven, E.E. (2026). **Lessons Learned from the Real-World Deployment of Multi-Sensor Fusion for Proactive Work Zone Safety Application**. In *Transportation Research Board Annual Meeting, Washington, DC, USA, Jan 11–15, 2026*. DOI: 10.48550/arXiv.2508.01599

[C.4] **Ahmad, M.U.**, Rahman, M., Sevim, A., Bodoh, D., Khan, S., Zhao, L., Huynh, N., and Ozguven, E.E. (2026). **Historical Prediction Attention Mechanism-Based Trajectory Forecasting for Proactive Work Zone Safety in a Digital Twin Environment**. In *Transportation Research Board Annual Meeting, Washington, DC, USA, Jan 11–15, 2026*. DOI: 10.48550/arXiv.2508.06544

[C.3] Saba, S.R., Khan, S., **Ahmad, M.U.**, Cao, J., Rahman, M., Zhao, L., Huynh, N., and Ozguven, E.E. (2026). **Infrastructure Sensor-Enabled Vehicle Data Generation Using Multi-Sensor Fusion for Proactive Safety Applications at Work Zone**. In *Transportation Research Board Annual Meeting, Washington, DC, USA, Jan 11–15, 2026*. DOI: 10.48550/arXiv.2509.25452

[C.2] Ibrahim, M.S., Rahman, S., Hasan, M.S., **Ahmad, M.U.**, and Abrar, A. (2022). **Flow-Based Path Planning for Multiple Homogenous UAVs for Outdoor Formation-Flying**. In *2022 7th International Conference on Mechanical Engineering and Robotics Research (ICMERR)*, pp. 18–26, IEEE. DOI: 10.1109/ICMERR56497.2022.10097797

[C.1] **Ahmad, M.U.**, Akib, A.A.R., Raihan, M.M.S., and Shams, A.B. (2022). **ABO3 Perovskites' Formability Prediction and Crystal Structure Classification Using Machine Learning**. In *International Conference on Innovations in Science, Engineering and Technology (ICISET)*, pp. 480–485, IEEE. DOI: 10.1109/ICISET54810.2022.9775906

Open-Source Contributions

- **GNSS-SDR [🌐]** *2026-*
  Software GNSS Receiver Contributor
  - Added a new PocketSDR signal source to GNSS-SDR source code.
  - Contributed to multiple data analysis tools of GNSS-SDR.

- **gr-pocketsdr [GitHub]** *2026-*
  GNU Radio OOT Module Repository Owner
  - Implemented a GNU Radio Out-of-Tree (OOT) block which enables interfacing PocketSDR front-ends to GNU Radio ecosystem.

- **OpenCAMS [GitHub]** *2025*
  SUMO-CARLA-OMNeT++ Co-Simulator Repository Owner
  - Developed a co-simulation environment that combines three major aspects of Intelligent Transportation System (ITS) - traffic simulation (SUMO), 3D sensor simulation (CARLA), and communications simulator (OMNeT++).
  - All source code is publicly available on GitHub.

- **MRTKLIB [GitHub]** *2026*
  A modernized implementation of RTKLIB with WebUI Contributor
  - Bug fixes.

Technology Demonstration

- **FMCSA Proactive Technology Training & Evaluation Workshop** *2025*
  University of Nebraska-Lincoln, Federal Motor Carrier Safety Administration Lincoln, NE
  - Demonstration of a proactive work zone warning system, built on roadside sensor fusion and trajectory prediction, to commercial motor vehicle drivers, construction workers, and DOT staff, including a hands-on demonstration through a driving simulator.

- **USDOT Future of Transportation Summit** *2024*
  US DOT Headquarters Washington DC
  - Demonstration of secure GNSS-based Autonomous Vehicle Navigation [🌐].

- **TraCR Annual Conference** *2024*
  National Center for Transportation Cybersecurity and Resiliency Greenville, SC
  - Demonstration of secure GNSS-based Autonomous Vehicle Navigation [🌐].

Teaching and Mentorship

- **Research Experiences for Undergraduates (REU)** *2026*
  Mentored undergraduate students under NSF-REU program University of Alabama
  - Craig Kudson
  - Rais Tefiku

Skills

- **Programming Languages:** Python, C, C++, Verilog
- **Database Systems:** SQLite, MongoDB, Redis
- **Data Science & Machine Learning:** Classical Machine Learning and Deep Learning, Computer Vision, Spatiotemporal Modeling
- **Development Technologies:** Linux, Git, GitHub/GitLab, Docker/Podman, Kubernetes, Proxmox
- **Specialized Simulation:** SUMO, CARLA, OMNeT++

Honors and Awards

- **Southwest Data Competition - University of Alabama - 2nd Place** *2026*
  Center for Regional and Rural Connected Communities (CR2C2) [🌐]
  - For the development of "Demand-Responsive Medical Transit Scheduling with Metaheuristic Optimization for Rural Healthcare Accessibility in Alabama Black Belt"

Leadership Experience

- **University of Alabama Student Chapter of the Institute of Transportation Engineers** *2025 - 2026*
  Vice President [🌐]
  - Organizing invited talks and programs in collaboration with Alabama Transportation Institute and student organizations.
- **IUT Mars Rover Team** *2021 - 2023*
  Team Lead: Software Subsystem [🌐]
  - Scored 10th place among 38 participating teams in European Rover Challenge (ERC) 2021.
  - Custom Linux distribution for Mars Rover.
  - Implementation of a ROS-based self-driving navigation stack based on visual odometry.
- **IEEE IUT Student Chapter** *2021 - 2023*
  Technical Officer [🌐]

Professional Memberships

- **Institute of Navigation (ION)**, Student member (Membership ID #MEM-336B63BB) *2026 - Present*
- **Institute of Electrical and Electronics Engineers**, Student member (Membership ID #101502991) *2026 - Present*
- **Institute of Transportation Engineers**, Student member (Membership ID #1114263) *2024 - Present*

Additional Information

Languages: English (Fluent), Bengali (Native)
Interests: Embedded and Linux systems, home server and self-hosting services, amateur radio.

References

1. **Dr. Mizanur Rahman**
   Associate Professor,
   University of Alabama
   Email: mizan.rahman@ua.edu
   Phone: +1 205 348 1717
   Relationship: PhD supervisor